

**Problem 7**

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**Problem 1** Evaluate the following integrals using symmetry arguments.

(a)  $\int_{-\pi/4}^{\pi/4} \sin(t) \, dt$

(b)  $\int_{-2}^2 (1 + x + 3x^7 - x^9) \, dx$

(c)  $\int_{-\pi}^{\pi} x \cos(x) \, dx$